

TEACHER VIVA & EVALUATION MASTER Q&A GUIDE

Easy Hinglish Answers (No Complex Math • Rule-Based Thresholds & Explanations)

Questions: English (Examiner Style)
Answers: Simple Hinglish (Easy to Speak)
Format: Crisp Bullet Points + "Why This Model"

MODULE 1: REPORT SECTION & AI VERIFICATION

Q1: "In the Report section, which AI is used, how does the algorithm verify fake vs genuine reports, which API is used, and why this model instead of heavy LLMs?"

Kaunsa AI Hai: Rule-Based NLP Agent aur In-Browser Computer Vision (Image Checker).

Kaise Check Karta Hai (Algorithm Rules):

- Fake Filter:** Agar message me 'test', 'prank', 'joke', 'haha' likha hai, ya 15 letter se chota hai, ya photo mobile ka screenshot hai → AI turant use **Fake / Spam** mark kar deta hai.
- Zameen Check (Terrain Physics):** Agar Cuttack jaisi flat zameen par koi 'Landslide' report karega, toh AI reject kar dega kyunki flat zameen par pahad nahi hote.
- Points Add Karna:** Bada descriptive message (+25), GPS location (+15), asli camera photo (+20), mausam match (+20) milte hain.
- Faisla:** **Score 70% se zyada → Genuine (Asli Report)**, **40% se 70% → Needs Review**, **40% se kam → Fake / Spam**.

Kaunsi APIs: Phone ka GPS (Geolocation API), Photo checker (Canvas API), aur local storage (IndexedDB).

🔍 Why This Model & Why Not Heavy LLMs (GPT-4 / Cloud CNN)?

"Sir, disaster me internet aur tower band ho jate hain. Agar hum heavy GPT-4 ya cloud model lagate, toh bina internet ke app fail ho jata aur mehenga bhi padta. Hamara model browser ke CPU par 1 millisecond me bina internet ke 100% offline chalta hai."

MODULE 2: LIVE ALERTS & TELEMETRY

Q2: "In the Live Alerts section, how are earthquakes and floods detected, what algorithm is used, and which APIs are integrated?"

Kaunsa System Hai: Telemetry Anomaly Detector (Live Sensor Processor).

Kaise Calculate Karta Hai (Algorithm Rules):

- Earthquake (Bhukamp):** USGS API se live Richter scale magnitude dekhta hai:
 - Agar magnitude **5.5 se zyada** hai → **Critical Red Alert** banta hai.
 - Agar magnitude **4.0 se 5.4 ke beech** hai → **Warning Yellow Alert** banta hai.
 - Kitne kilometer tak asar hoga (Impact Radius) nikal kar map par circle bana deta hai.
- Flood (Baadh) & Toofan:** Open-Meteo satellite se barish aur hawa check karta hai:
 - Agar barish **5 mm/hour se zyada** hai → **Flash Flood Warning** banti hai.
 - Agar hawa ki speed **35 km/hour se zyada** hai → **Cyclone / Gale Alert** banta hai.

Kaunsi APIs: USGS Global Earthquake API aur Open-Meteo Weather API.

🔍 Why These APIs & Why Not Paid Ones?

"Sir, USGS aur Open-Meteo government aur open-science live feeds hain. Yeh 100% free hain, inme koi billing limit nahi hoti aur asli physical sensors ka data deti hain jisse jhootha alerts nahi aate."

MODULE 3: RISK PROBABILITY & RADAR

Q3: "In the Dashboard & Risk Forecast section, how is disaster risk probability calculated, and why use this model instead of simple percentages?"

Kaunsa Model Hai: Bayesian Bell Curve (Risk Graph) aur 6-Axis Radar Chart.

Kaise Calculate Karta Hai (Algorithm Rules):

- Bell Curve Graph:** Live mausam ke hisaab se curve ka peak aage-piche shift hota hai:
 - Normal mausam me curve **Low Risk (20-25%)** par rehta hai.
 - Jaise hi barish aur hawa badhti hai, curve ka peak **High Risk (75-85%)** par shift ho jata hai.
- Radar Chart:** Ek sath 6 khatron (Flood, Cyclone, Heatwave, Earthquake, Drought, Tsunami) ka live 0 se 100% meter ek hi hexagon graph me dikhata hai.

Kaunsi API / Tool: Open-Meteo Weather API aur Recharts visualization library.

🔍 Why This Model & Why Not Simple Static Percentages?

"Sir, sirf '60% barish hogi' bolne se disaster ka exact khatra samajh nahi aata. Bell Curve graph citizen aur rescue teams ko poora probability spread aur sensor ka confidence dikhata hai."

MODULE 4: DISASTER MAP & SAFE ROUTING

Q4: "In the Map & Navigation section, how does shortest safe path routing work, which algorithm is used, and what happens when internet goes down?"

Kaunsa Algorithm Hai:

1. **Shortest Safe Path ke liye: Dijkstra / OSRM Algorithm** (ye road network par flooded aur blocked raste ko chhodkar sabse chhota aur safe rasta nikalta hai).
2. **Do Points ke beech Distance ke liye: Haversine Algorithm** (GPS coordinates se exact kilometer distance calculate karta hai).
3. **Offline Mode ke liye:** Agar internet band ho, toh hamara offline curve formula 32 km/h emergency speed ke hisaab se turn-by-turn rasta aur pahuche ka time (ETA) nikal deta hai.

Kaunsi APIs: Project OSRM Routing Machine, Komoot Photon OpenStreetMap API, aur Leaflet Map.

Why Leaflet & OSRM Instead of Google Maps API?

"Sir, Google Maps API mehenge charges lagati hai aur offline caching allow nahi karti. Leaflet aur OSRM 100% free hain, open-source hain aur flooded khatron ko bypass karke offline safe path nikalte hain."

MODULE 5: NEARBY HELP & 1-TAP SOS

Q5: "In the Emergency Help section, how are nearby hospitals and shelters found, and how does 1-tap SOS work?"

Kaunsa Algorithm Hai: Bounding Box Search + Proximity Sorting.

User ke GPS se 15-45km radius me hospitals, police station, fire brigade, aur cyclone shelter dhoondh kar sabse paas wale ko top par sort karta hai.

1-Tap SOS Kaise Kaam Karta Hai:

Direct phone ka dialer open karta hai bina kisi typing ke:

- Police: 112 • Ambulance: 108 • Fire Brigade: 101

Sath hi WhatsApp/SMS par exact GPS location broadcast kar deta hai.

Kaunsi APIs: Komoot Photon OSM API, Tel URI Protocol, aur Web Share API.

Why Direct Hardware Protocol?

"Sir, emergency me victim ke paas type karne ka time nahi hota. 1-Tap direct phone call lagata hai aur bina internet ke location share kar deta hai."

MODULE 6: VOLUNTEER DASHBOARD & TRIAGE

Q6: "In the Volunteer Dashboard, how are genuine incidents prioritized and dispatched to volunteers?"

Kaunsa Algorithm Hai: Priority Triage Sorting + 6-Step Mission Flow.

1. Sabhi fake aur unverified reports ko hata deta hai (sirf `isGenuine === true` dikhti hain).
2. Priority ranking: **Critical (Jaan ka khatra) > Medium > Low.**
3. Same priority me jo incident volunteer ke sabse paas hai, wo sabse upar aata hai.
4. **6-Step Flow:** Report aayi → Claim kiya → EnRoute (Raste me) → OnScene (Pahucha) → Madad di → Resolved (Complete).

Why This Model?

"Sir, volunteers ka time bohot keemti hota hai. Ye system fake calls ko rokta hai aur ensure karta hai ki duplicate rescue teams ek hi jagah na jayein."

MODULE 7: OFFLINE PERSISTENCE & SYNC

Q7: "In the Offline Data section, how does the app store data during network blackouts and sync when internet comes back?"

Kaunsa Architecture Hai: IndexedDB Local Database + Offline Sync Queue.

1. Internet na hone par report phone ke IndexedDB me PendingSync tag ke sath save ho jaati hai.
2. Jaise hi internet aata hai, hamara OfflineSyncManager background me chupke se reports server par sync karke status Verified kar deta hai.

Kaunsi Browser APIs: IndexedDB v2 (5 Stores) aur Vite PWA Service Worker.

Why IndexedDB Why Not LocalStorage?

"Sir, LocalStorage sirf 5MB data rakh sakta hai aur phone ko hang kar deta hai. IndexedDB 500MB+ data aur photos ko bina kisi UI lag ke offline store kar sakta hai."

Section	Kaunsa Model Hai?	Kaunsa Algorithm Chal Raha Hai?	Why This Model? (1 Line)
1. Report Section	Bayesian NLP + Image Checker	Points Scoring ($\geq 70\%$ Genuine, $< 40\%$ Spam)	Bina internet ke phone me 1ms me chalta hai.
2. Live Alerts	Earthquake & Flood Detector	USGS Richter (≥ 5.5 Critical) + Rain ($\geq 5\text{mm}$ Flood)	Real government sensor data, zero false alarms.
3. Risk Forecast	Bayesian Bell Curve + Radar	Live Weather Shift (Low 25% $\rightarrow$ High 85%)	Static number ke badle complete risk spread dikhata hai.
4. Disaster Map	OSRM Dijkstra + Sinusoidal	Haversine Distance + Flooded Obstacle Bypass	100% Free, open-source aur offline safe rasta deta hai.
5. Volunteer Desk	Priority Triage + 6-Step Flow	Critical > Medium > Nearest Proximity	Fake calls rokta hai aur duplicate rescue bachata hai.
6. Offline Sync	IndexedDB + Sync Queue	FIFO Auto Reconnection Sync	500MB+ offline storage deta hai bina phone hang kiye.